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Title:

Making an automated brightness adjustable light using Arduino.

Objective:

To develop a brightness adjustable light to provide the user with a versatile lighting solution including energy savings, extension of lifetime of bulb, providing relaxing and calming environment according to customer needs.

Introduction:

Energy efficiency and cost reduction are two major important terms that very much related to our ongoing life. With the rise in energy costs and growing concerns about the environment, people are looking for ways to reduce their energy consumption. Developing an efficient brightness light can be an alternative to low efficient normal LED light which will provide energy efficiency, health promoting, flexible and aesthetically appealing lighting solutions.

Our projects aim is to design and implement a smart lighting system that adjusts its brightness based on the amounts of sunlight in the surroundings environment. An ambient light sensor is used to detect the level of natural light and adjust the brightness of the artificial light source accordingly. The project involves the selection of appropriate components, programming of a microcontroller and testing of the system under different lighting conditions.

The ultimate goal is to create a cost effective, user friendly and sustainable lighting solutions that benefits both the environment and the end user.

Components:

- Light-dependent resistor (LDR)
- Microcontroller (such as an Arduino Uno)
- LED lights
- Power Supply
- Resistors
- Breadboard
- Jumper wires

Tools:

- Soldering iron and solder
- Multimeter
- Computer with software for programming the microcontroller

Generate product concepts:

CONCEPT SELECTION:

Finalized concept screening

Selection Criteria	**CONCEPT A** (Manual Brightness Adjustable Light)	**CONCEPT B** (Automated Brightness Adjustable Light)	**CONCEPT C** (Voice Control IoT Light)	**CONCEPT D** (Normal LED Light Bulb) **Reference**
Durability	+	+	+	0
Cost	-	+	-	-
Energy efficiency	+	+	-	-
Sustainability	+	+	0	0
Ease of manufacture	+	+	-	-
Brightness	+	+	+	+
Sum +'s Sum 0's Sum –'s	5 0 1	5 0 0	2 1 3	1 2 3
Net score Rank Continue?	4 2 Yes	5 1 Yes	-1 3 No	-2 4 No

Finalized concept scoring

Selection Criteria	Weight	Concept A		Concept B	
		Rating	Weighted score	Rating	Weighted score
Durability	15%	5	0.75	5	0.75
Cost	15%	2	0.30	2	0.30
Energy efficiency	20%	3	0.60	4	0.80
Sustainability	15%	3	0.45	5	0.75
Brightness	25%	4	1.00	4	1.00
Ease of manufacture	10%	5	0.50	5	0.50
	Total score rank		3.60		4.10
	Continue?	No		Develop	

Product Architecture:

Product architecture is the assignment of the functional elements of a product to the physical building blocks of the product. In considering the correct architectural model, we the team members asked ourselves:

- How would the architecture of the product impact their ability to offer product variety?
- What would be the cost implications of different product architectures?
- How would the architecture of the product impact their ability to complete the design within 12 months?
- How would the architecture of the product influence their ability to manage the development process?

After a fair bit of discussion, we decided to stick with integrated architecture since it gave us the maximum amount of flexibility in the production process without losing its essence of it. Some other factors that were also kept in mind for selecting the right architecture were:

• Upgrade

• add-ons

• Adaptation

• Wear

• Consumption

• Flexibility in use

• Reuse

We divided all the functional elements into different modules and later arranged them into different chunks:

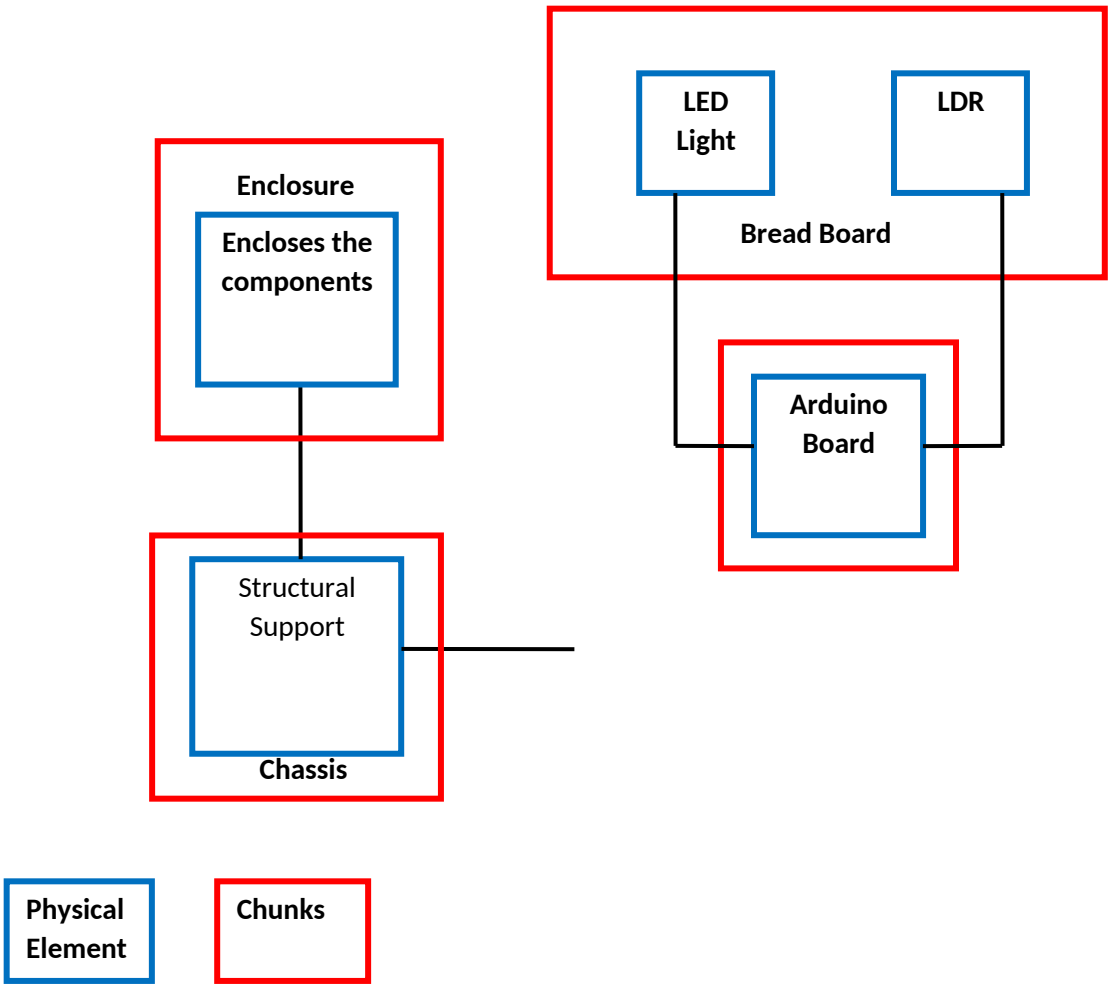


Fig: ***Cluster of elements into different chunks***

Bill of materials:

Components name	Quantity	Unit cost (Tk)	Total cost
Arduino uno board	1 pc	850 tk	850 tk
Light dependent resistor	2 pc	30 tk	60 tk
Breadboard	1 pc	150 tk	150 tk
LED light	3 pc	5 tk	15 tk
Power supply	1 pc	50 tk	50 tk
Resistor	2 pc	2 tk	4 tk
Jumper wires	6 pc	5 tk	30 tk
• Soldering iron and solder • Computer with software		100 tk	100 tk
Total direct cost Overhead charges **Total cost**			1259 tk 220 tk 1479 tk

Table: Bill of materials (BOM) table

Fabrication

- We begin the fabrication process by purchasing an Arduino Uno board, breadboard, LED sensor, plywood, etc. at a low price.
- Next, we cut the plywood into two equal pieces measuring 30cm x 20cm.
- Then, we position the breadboard and Arduino in a suitable location on the plywood.
- Using glue, we attach the components to the plywood.
- Finally, we connect all the pieces of plywood to create the proto

Assembly

Here are the steps to make the brightness-adjustable light:

- At first, we connect the LDR to the microcontroller using jumper wires. The LDR should have two pins - one for power and one for the signal. Connect the power pin to 5V and the signal pin to any analog input on the microcontroller.
- Then we connect the LED lights to the microcontroller using resistors. Calculating the appropriate resistor value using Ohm's law based on the LED's voltage and current ratings and connect the LED anode (positive leg) to the resistor and then to the microcontroller's digital pin. Lastly, we connect the LED cathode (negative leg) to ground.

- After the joining procedure we program the microcontroller to read the LDR's data and adjust the LED brightness accordingly. For this we use a programming language such as C++ + a development environment such as the Arduino IDE. Write code that reads the sensor's analog output and maps it to the appropriate LED brightness level.
- After that we test the system by placing it in different lighting conditions. The LED brightness should adjust automatically based on the amount of light in the surrounding environment.

  .

With these steps, we should be able to create a brightness-adjustable light that controls the brightness according to the light of the surrounding environment.

Prototype:

Product Scetch:

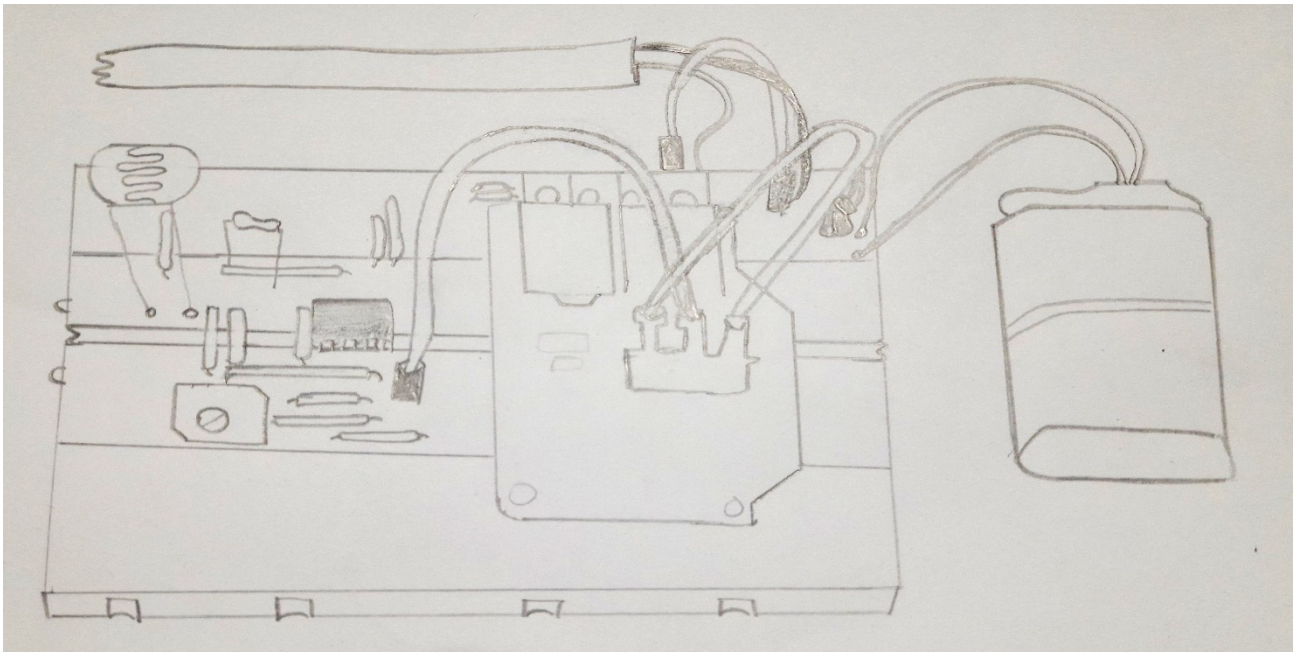
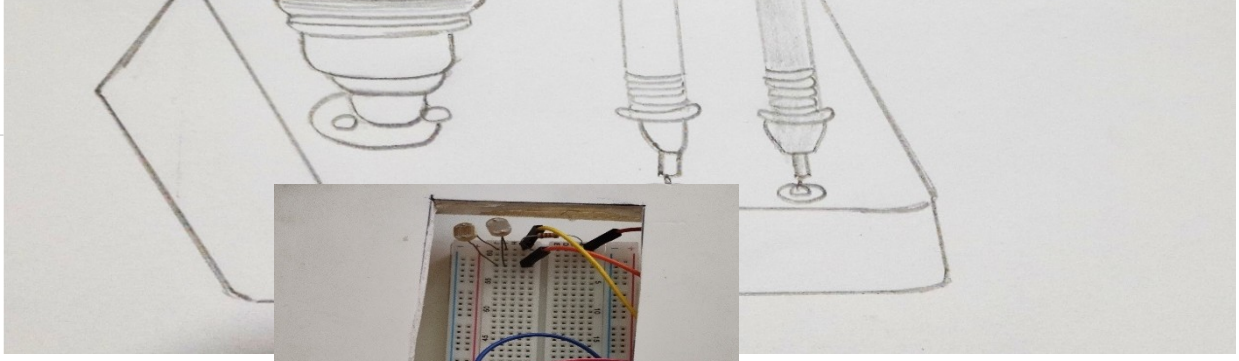
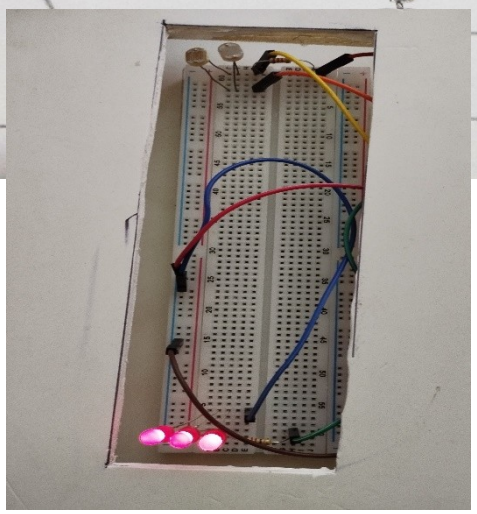


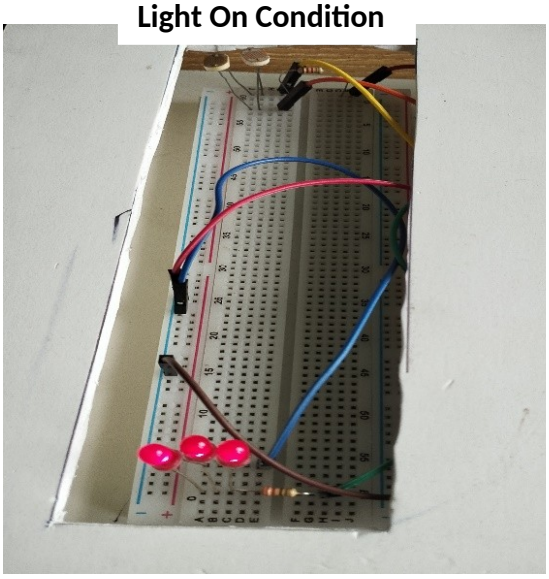
Fig: Internal design





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Light On Condition



Light Off Condition

Assembly:

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With these steps, we should be able to create a brightness-adjustable light that controls the brightness according to the light of the surrounding environment.

Coding:

```
int potpin=A0;
int ledpin=11;
int val=0;
void setup()
{
pinMode(ledpin,OUTPUT);
Serial.begin(9600);
}
void loop()
{
val=analogRead(potpin);
Serial.println(val);
analogWrite(ledpin,val);
delay(10);
}
```

Advantages:

There are several reasons why having a brightness-adjustable light can be beneficial:

- **Comfort:** Adjusting the brightness of a light can help to create a comfortable environment. For example, if the light is too bright, it can cause eye strain and headaches, whereas if the light is too dim, it can make it difficult to see properly. By adjusting the brightness to a comfortable level, you can reduce the risk of these issues.
- **Energy savings:** If you use a light that is brighter than necessary, it can waste energy and increase your electricity bill. By adjusting the brightness to a lower level, you can save energy and money.
- **Mood and ambiance:** A light with adjustable brightness can help you to set the mood and ambiance of a room. For example, a dimmer light can create a more relaxing atmosphere, whereas a brighter light can create a more energetic atmosphere.
- **Task-specific lighting:** Depending on the task at hand, you may need different levels of lighting. For example, if you're reading a book, you may need brighter lighting to see the

text clearly. If you're watching a movie, you may want a dimmer light to create a more immersive experience.

- **Longer Life Span:** Dimming the lights can extend the life span of the bulbs or LED strips, as they are not operating at full power. This reduces the frequency of replacement and hence, the overall cost.
- **Comparison to Market Prices:** There is no existing product that utilizes a microcontroller in the same way as this brightness-adjustable light, which adjusts its brightness based on the surrounding environment. Therefore, this innovative product would be quite exceptional and allow us to set a price that suits us. This product has the potential to boost our business.
- **Cost Savings:** By making a brightness-adjustable light that controls the brightness according to the light of the surrounding environment, energy savings can be achieved by using the minimum amount of light required. This can result in significant cost savings on electricity bills in the long run.

Overall, having a brightness-adjustable light can help to create a more comfortable, energy-efficient, and versatile lighting environment.

Uniqueness:

1. The cost is affordable.
2. Automated brightness adjustability.
3. It is very much compatible with smart home system.
4. Brightness level and color temperature of the lights have higher health benefits.

Area of improvements:

- ✓ Brightness adjustable lights can be more expensive than traditional lights, which may make them less accessible for some consumers. However, as the technologies continues to improve, the cost is likely to decrease over time.
- ✓ While some brightness adjustable lights allow for color temperature adjustment, the color accuracy may not always be perfect.
- ✓ Brightness adjustable light is more complex to install and set up than traditional lights. So manufacturers have the chance to simplify the installation process for easy accessible to all customer.
- ✓ There are also scopes for improving the reliability as well.

Comparative Cost Analysis:

Let's compare our automated brightness adjustable light bulb (Product A) with a Normal LED Light Bulb (Product B) and an Voice-over-IoT light bulb (Product C) and a Manual Brightness Adjustable Light Bulb (Product D) in terms of cost. Here's a comparative cost analysis:

Product A (Automated Brightness Adjustable Light Bulb):

- Price: $10
- Key Features: Brightness adjustable with manual control, standard LED technology
- Total Cost of Ownership: No significant additional costs

Product B (Normal LED Light Bulb):

- Price: $5
- Key Features: Fixed brightness, standard LED technology
- Total Cost of Ownership: No significant additional costs

Product C (Overpriced Voice-over-IoT Light Bulb):

- Price: $30
- Key Features: Brightness adjustable with voice control through IoT connectivity, premium LED technology
- Total Cost of Ownership: Potential additional costs related to IoT connectivity, compatibility, and maintenance

Product D (Manual Brightness Adjustable Light Bulb):

- Price: $15
- Key Features: Brightness adjustable with manual control, standard LED technology
- Total Cost of Ownership: No significant additional costs

Cost per Unit of Functionality:

- Product A: $10 (Adjustable brightness)
- Product B: $5 (Fixed brightness)
- Product C: $30 (Adjustable brightness + IoT connectivity)
- Product D: $15 (Adjustable brightness)

Our innovative brightness-adjustable light bulb stands out in the market due to its unique features. Unlike existing products, it utilizes a microcontroller to automatically adjust brightness based on

the surrounding environment. This sets it apart from manually controlled lights and expensive voice-over-IoT alternatives. Our light bulb is cost-effective, energy-efficient, and simpler to manufacture, making it more affordable for the general population. With these advantages, our product has the potential to boost our business, compete with other LED bulbs, and dominate the market in the future.

Prospects of the product in Bangladesh:

The prospect of automated brightness adjustable lights is promising, as they offer a number of benefits over traditional solutions. Thus, it depends upon various factors such as:

Feasibility:

The feasibility of brightness adjustable light system in Bangladesh would depend on several factors, such as the availability of the necessary technology, the cost of implementation, and the level of demand from consumers.

The necessary sensors and lighting systems are readily available on the global market, so it is likely that they could be sourced in Bangladesh as well. However, the cost of the components and the overall installation cost could be a challenge for the product particularly for household or the business with limited budgets. Thus, energy costs and environmental factors are in rise nowadays. So, considering this there may be increasing demand for energy efficient lighting solutions in future.

Available manufacturers:

Quality and availability of the product may vary depending on the manufacturers and available components. The internal mechanisms for the product are very complex. So it’s not accessible for local manufacturers. There need trained professional who have the knowledge of programming and electrical components to manufacture the product.

Market Outlook:

Market outlook of any product refers to the anticipated performance of the product in the market over a specified time period. It depends upon various factors including target market, market trends, consumer behavior, economic condition, technological advancement etc. The market outlook helps business to identify potential opportunities and challenges in the market, and adjust their strategies accordingly.

As our product is in primary stage thus, we need to design a market outlook for the product based on the key factors to develop market strategies, set pricing policies, make investment decisions and determine the production level successfully.

Discussion:

We began our product design and development journey by identifying the demands of our target customers. Specifically, we looked at how we might ensure energy savings, comfortable brightness at rooms with automated brightness adjustable light. After much observations we have developed a prototype of automated brightness adjustable light which will adjust its brightness according to surrounding illumination. It will reduce the energy cost. Moreover, it will ensure comfort and healthy life of the users. We used some small LED lights in our prototype because of the unavailability of larger LED lights in the local markets and to reduce our cost. Though the cost of our product may seem higher than other lights but it will reduce on mass production.

There are many unique features in our product which will attract the customers. Such as it adjusts its brightness according to surrounding illuminance. So, it will save energy and cost. There is no existing product in the market with this feature. Moreover, it will ensure the comfort and good health of the users.

High use of energy needs increased rate of fuel burning. Excessive fuel burning causes climate change and global warming. So, to save our earth it is must to reduce the consumption of energy as much less as possible. Our brightness adjustable light is a great solution of this. Moreover, by adjusting brightness it will give a suitable environment in rooms which will ensure good mental health and increase productivity.

Conclusion:

The ultimate purpose of this entire process was to develop a product which will reduce the cost of energy and ensure a healthy human life. Also, it should be commercially viable. In order to achieve this goal a systematic strategy with well planning was implemented which provided us with a definitive outcome in the end. Out team members have put up their best efforts to complete this project.